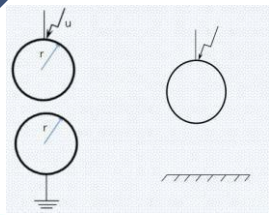
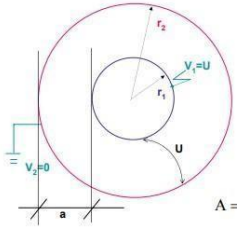
EE4047 – High Voltage Techniques I

New Topic:

Electrical Breakdown in Gases

Week 14





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02.06.2026

OUTLINE

2 Multilayer Dielectric Concentric Spheres

OUTLINE

1 Some Basic Concepts

2 Introduction to Breakdown in Gases

3 Ionization Types in Gases

4 Discharge theorems

1 SOME BASIC CONCEPTS

Insulation

To prevent electrical conduction between points with different potentials

Insulator

Material used for obtaining electrical insulation in the insulation system

- Gas
- Liquid
- Solid
- Vacuum

1 SOME BASIC CONCEPTS

Electrical discharge

Any dielectric (or insulator) can withstand a voltage applied below a critical value.

If applied voltage across dielectric exceeds this critical value, a discharge occurs in the insulator medium.

1 SOME BASIC CONCEPTS

Electrical discharge

There are different types of electrical discharge.

- **Breakdown:** Complete electrical discharge through the insulator
- **Flashover:** Complete electrical discharge jumping around the insulator
- **Partial discharge:** Partial breakdown on conductors at points with the highest electrical stress

2 INTRODUCTION TO BREAKDOWN IN GASES

Gases as insulating medium is often preferred in high voltage technique application because of their self-restoring capability after a breakdown.

The most common gases are air and SF_6.

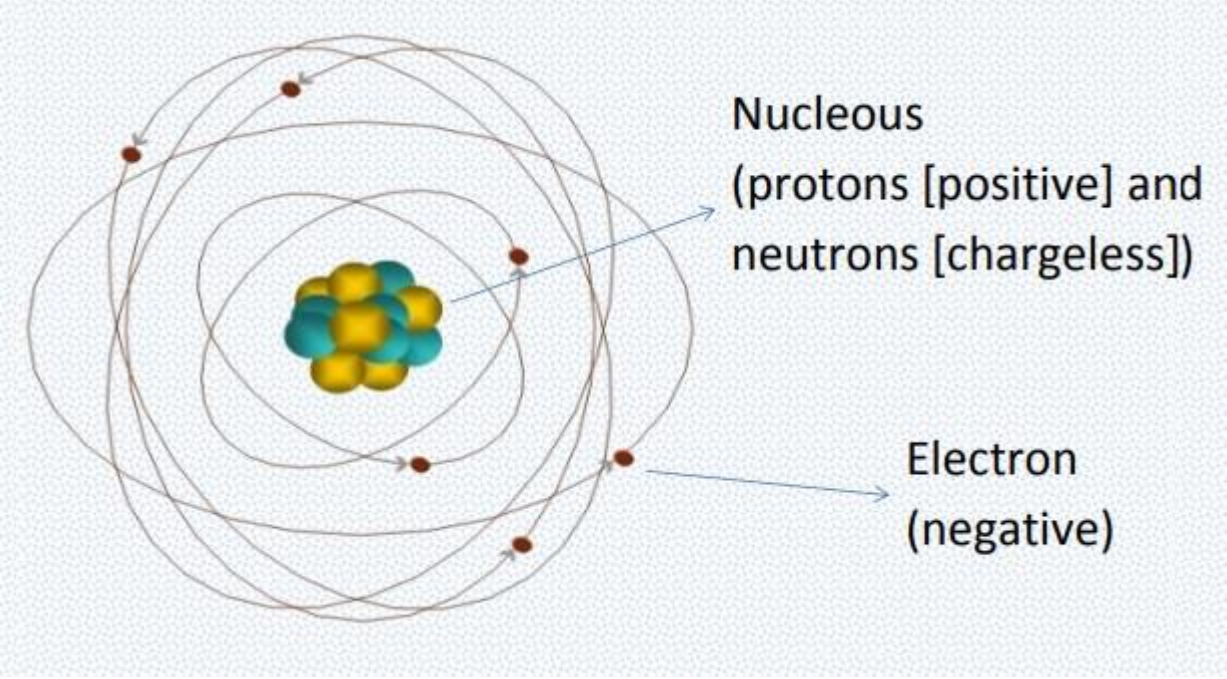
Discharge in gases

Generally, a neutral gas does not conduct electricty when it is conserved from external factors. However, application of electric field can lead the gas to lose its insulating properties at a critical value.

This conduction in dielectric is called ELECTRICAL DISCHARGE. Electrical discharge occurs in the presence of charged particles which are electrons (negatively charged particles) and ions (negatively or positively charged atoms or molecules).

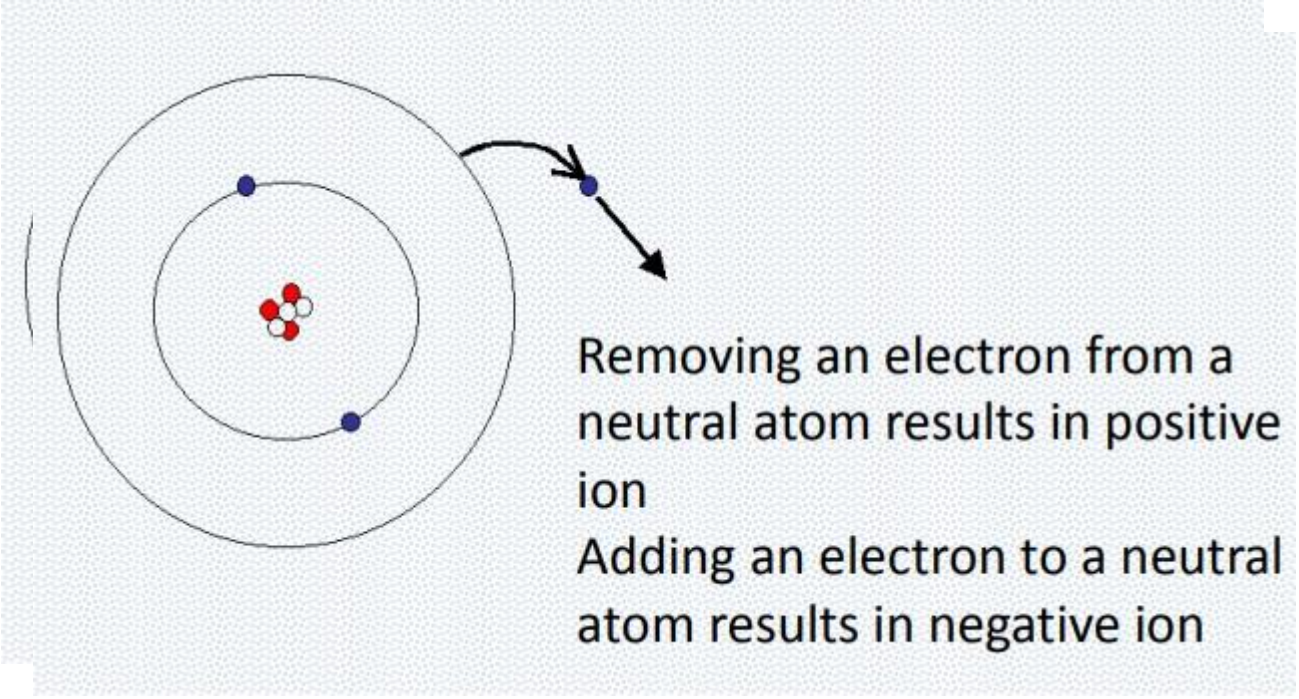
2 INTRODUCTION TO BREAKDOWN IN GASES

Structure and excitation of atom and ionization



2 INTRODUCTION TO BREAKDOWN IN GASES

Structure and excitation of atom and ionization



2 INTRODUCTION TO BREAKDOWN IN GASES

Energy on a charge

$$F = q\,E$$

$$E = \frac{U}{d}$$

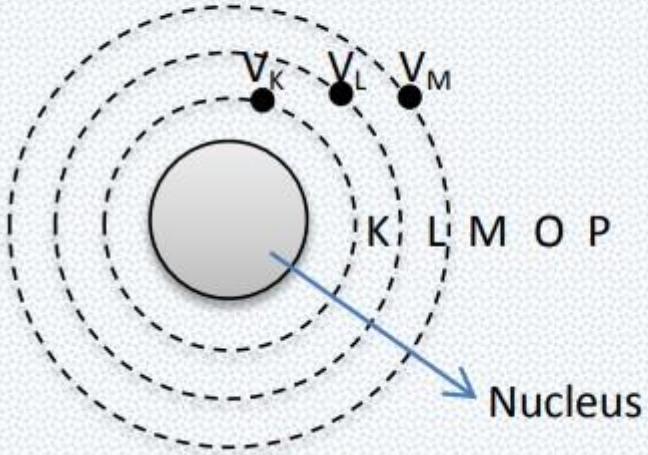
$$F = q\frac{U}{d}$$

$$Fd = qU = W\ (energy, work)$$

2 INTRODUCTION TO BREAKDOWN IN GASES

Excitation energy

$$W_e = q\,U_e$$



$$W_e = q\,(V_K - V_L)$$

When a voltage between two orbits is applied to an electron, electron jumps to the one orbit level up.

INTRODUCTION TO BREAKDOWN IN GASES

Photon energy

The electron stays at this new orbit level for about $10^{-8} - 10^{-9}$ seconds and it return to its original position.

The electron emits energy as photon (beam) energy.

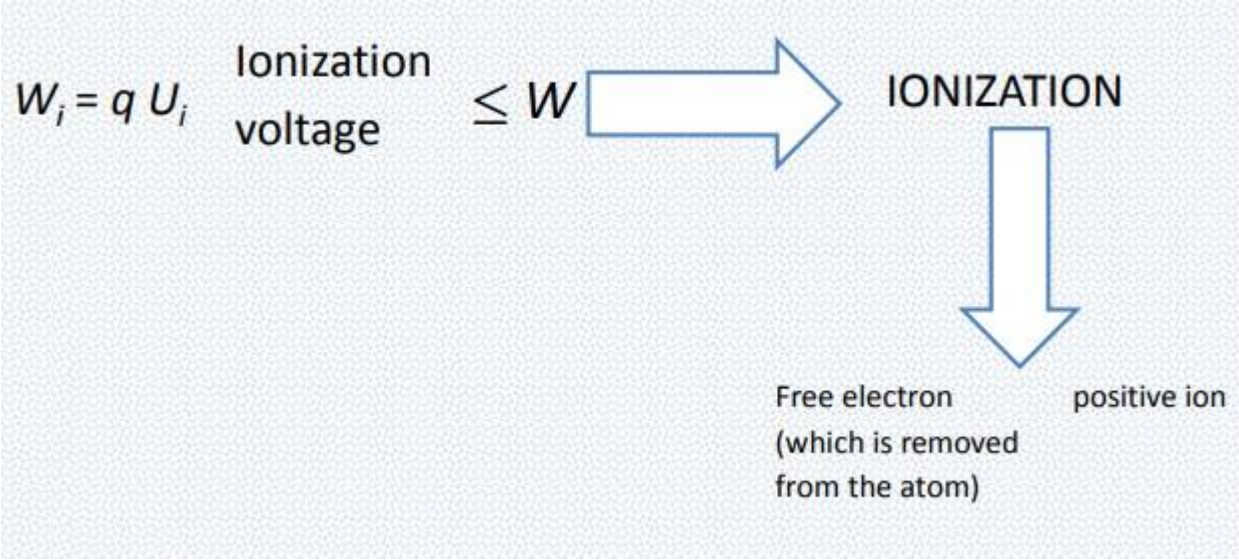
$W_e = qU_e = h\,f_e$ Photon energy (quantum of energy)

h: Planck's constant; f_e: frequency of the radiation

3 IONIZATION TYPES IN GASES

Ionization

Removing an electron from an atom or a molecule is called ionization.



3 IONIZATION TYPES IN GASES

1. Ionization by collision

Gas atoms and molecules collect kinetic energy by moving. When atoms or molecules collide, they transfer their energies to each other.

$$E_k = \frac{1}{2}mv^2 : \text{kinetic energy}$$

$$W_i \leq \frac{1}{2}mv^2 \rightarrow ionisation\ by\ collision$$

When kinetic energy of an electron exceeds the ionization energy of the atom or molecule (the energy needed to be given to an atom or a molecule in order to remove an electron), ionization occurs. Sometimes ionization occurs progressively.

3 IONIZATION TYPES IN GASES

2. Photo-ionization

A molecule in the ground state can be ionized by a photon having frequency f provided that the quantum energy emitted h (by an electron jumping from one orbit to another), is greater than the ionization energy of the molecule (gas gains photon's energy).

$$W_p = h\, f_p : \text{Photon energy}$$

h: Planck's constant = 6.625×10^{-34} Joule.s; f_p: frequency of photon

3 IONIZATION TYPES IN GASES

2. Photo-ionization

$f = \frac{v}{\lambda}$, $\lambda \downarrow \rightarrow f \uparrow \rightarrow W = hf \uparrow$ $\lambda = \frac{c}{f}$

(v: speed & λ: wavelength)

$$W = hf \geq W_i \rightarrow photoionisation$$

Photon with a small wavelength (λ) or high frequency (f) can supply enough energy for ionization.

3 IONIZATION TYPES IN GASES

3. Thermal ionization

1. Gas molecules move faster when heated.

$$E_k = \frac{1}{2}mv^2 : \text{kinetic energy increases}$$

$$W_i \leq \frac{1}{2}mv^2 \rightarrow ionisation\ by\ collision$$

3.2. Gas emits light when heated

$$W = h\,f \geq W_i \rightarrow photoionisation$$

3 IONIZATION TYPES IN GASES

4. Surface ionization

1. Electron Bombardment:

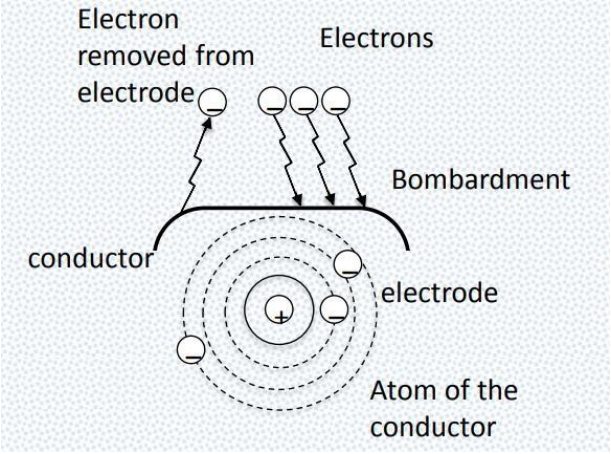
When electrons or positive ions impinge to the electrode, they will remove electron.

Minimum energy that has to be supplied to the conductor atom or molecule is called <u>output energy</u>

$$W_o \ll W_i = 2 - 25\ eV$$

W_o: 0.1 – 1 eV, Output energy of the electrode

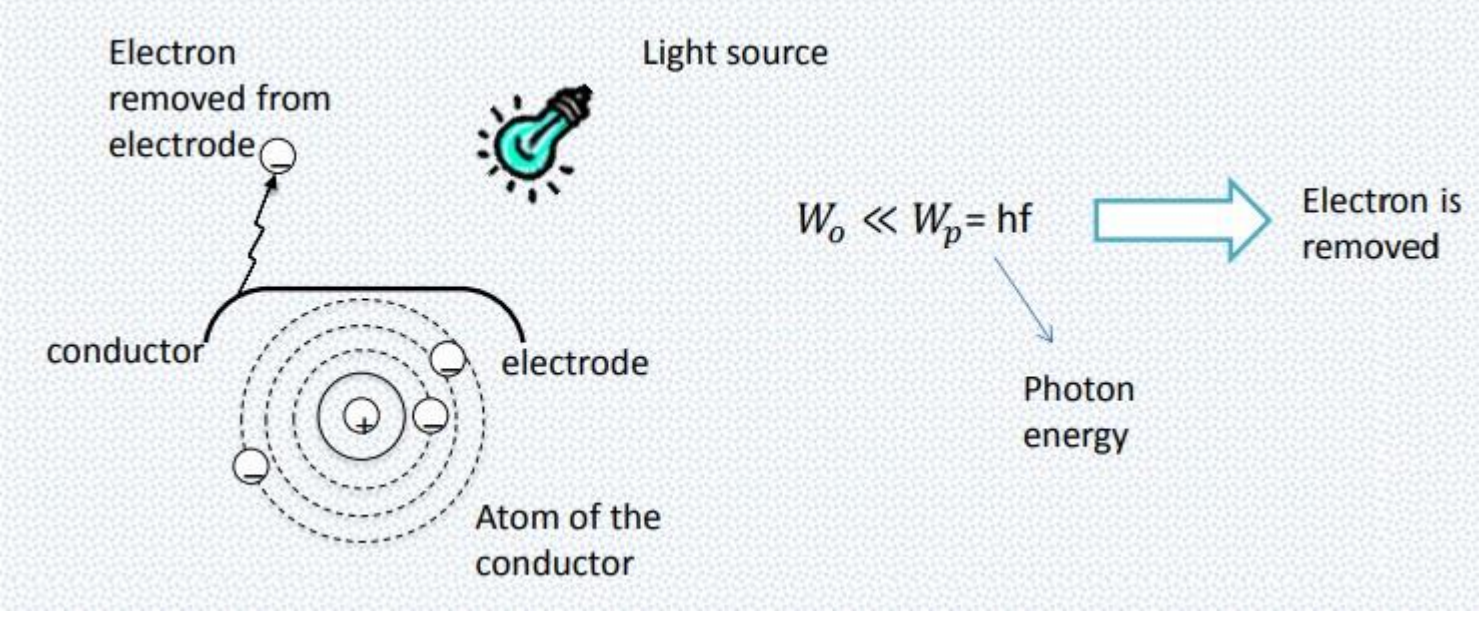
W_i: ionization energy of the gas



3 IONIZATION TYPES IN GASES

4. Surface ionization

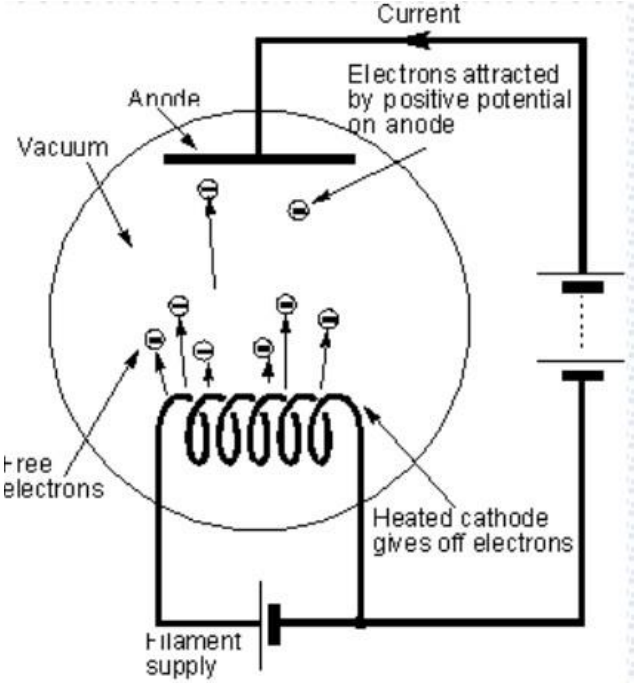
4.2. Surface ionization by light:



3 IONIZATION TYPES IN GASES

4. Surface ionization

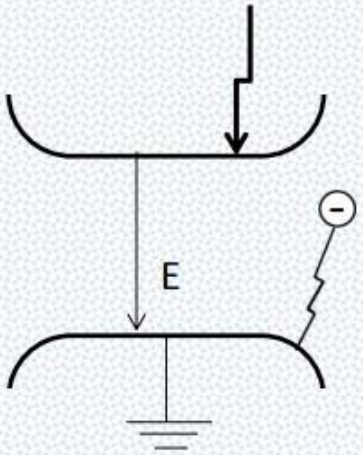
4.3. Surface ionization by heat:



3 IONIZATION TYPES IN GASES

4. Surface ionization

4.4. Surface ionization by field effect:



F = q E

E = U/d

F = q U/d

$W_{o,atom} = 1 - 5\ eV$

$W_{i,gas} = 2 - 25\ eV$

$W_o < W_{i,gas}$

it is easier to remove an electron from metal rather than gas!!!

1 DISCHARGE THEOREMS

1. Townsend Theorem

Electron Avalanche mechanism:

Suppose a free electron exists (due to cosmic radiation or some other external effect) in a gas where an electric field exists.

If E is sufficiently high, ionization by collision is very possible.

If we apply E, free electrons are accelerated

- Kinetic energy
- in eV
- move towards the anode

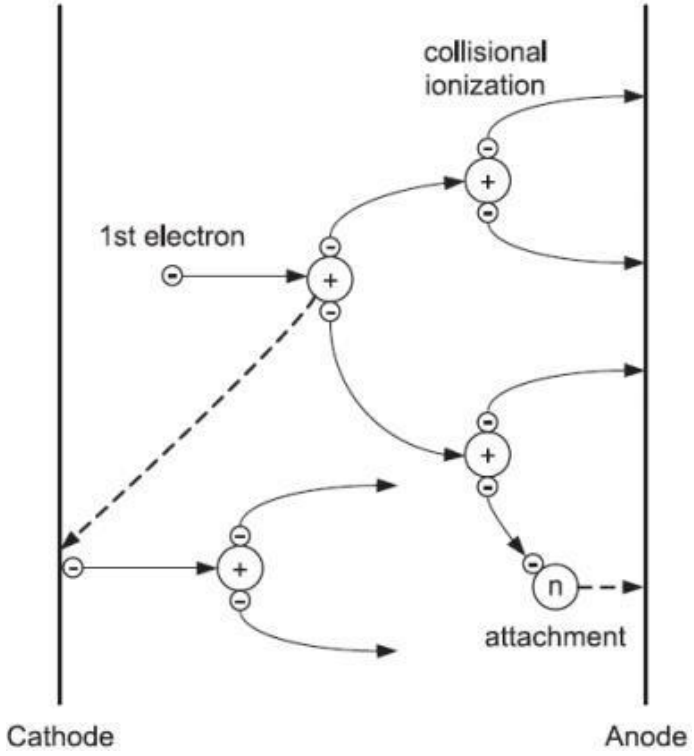
1 DISCHARGE THEOREMS

1. Townsend Theorem

Collision:

This process is cumulative.

The number of electrons will go on increasing as they continue to move under the action of E.



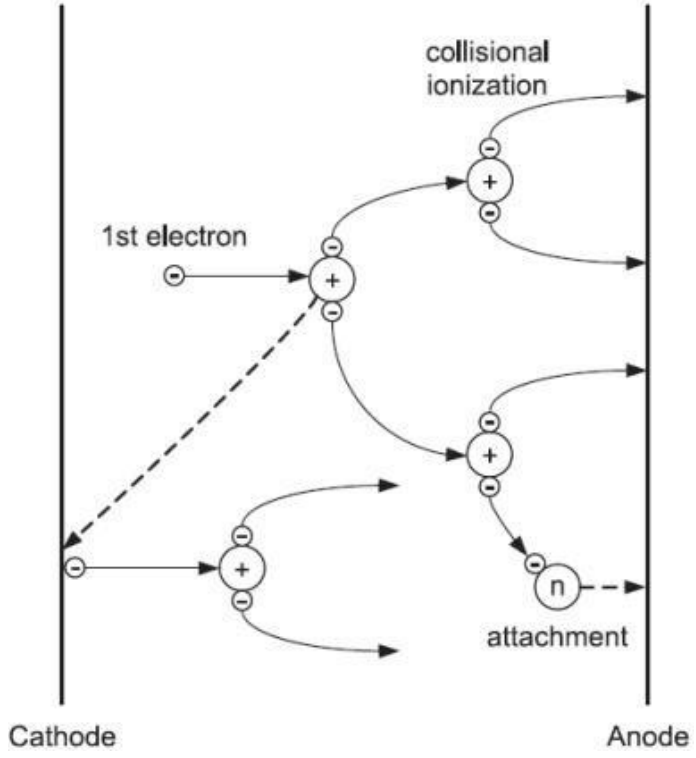
1 DISCHARGE THEOREMS

1. Townsend Theorem

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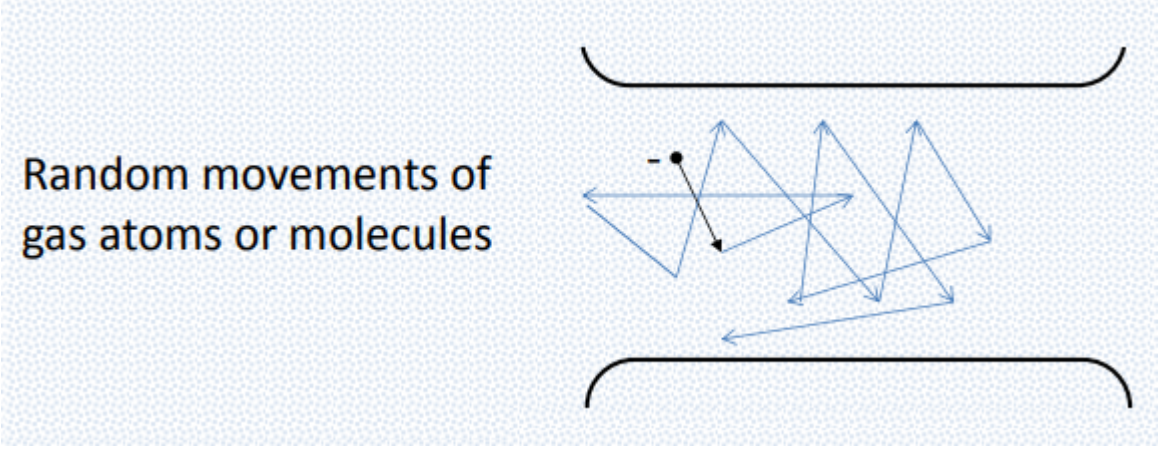


1 DISCHARGE THEOREMS

1. Townsend Theorem

Free Path law:

Free Path (λ) is the distance traveled by a particle between two collisions.



1 DISCHARGE THEOREMS

1. Townsend Theorem

Mean Free Path ($\lambda_{ave} = \lambda$)

It is the arithmetic average distance travelled by a moving particle between successive impacts.
or
Arithmetic average distance travelled by more than one particle in a certain period of time.

$$\bar{\lambda} = \lambda_{ave} = \frac{1}{n}(\lambda_1 + \lambda_2 + \cdots + \lambda_n)$$

n: number of collision, or number of free paths

1 DISCHARGE THEOREMS

1. Townsend Theorem

Free Path law:

Probability of a particle to travel a distance equal or greater than λ

$$f(\lambda) = P(\lambda) = e^{-\frac{\lambda}{\lambda_{ave}}}$$

Probability of $\lambda \geq 0$ $\quad e^{-\frac{0}{\lambda_{ave}}} = e^{0} = 1 \rightarrow 100\%$

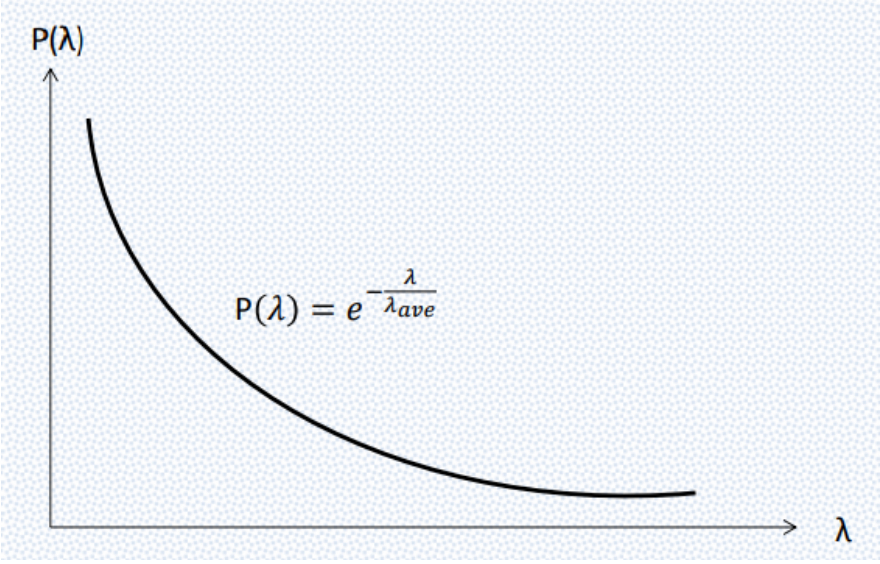
Probability of $\lambda \geq \lambda_{ave}$ $\quad e^{-\frac{\lambda_{ave}}{\lambda_{ave}}} = e^{-1} = \frac{1}{e} \rightarrow 37\%$

Probability of $\lambda \geq 3\lambda_{ave}$ $\quad e^{-\frac{3\lambda_{ave}}{\lambda_{ave}}} = e^{-3} = \frac{1}{e^{3}} \rightarrow 5\%$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Probability function:



DISCHARGE THEOREMS

1. Townsend Theorem

- Number of initial particles: n_0
- Number of non-colliding particles: n_λ

$$n_\lambda = n_0\, e^{-\frac{\lambda}{\lambda_{ave}}}$$

$$\begin{matrix}\text{Number of} \\ \text{total particles}\end{matrix} = \begin{matrix}\text{number of} \\ \text{colliding particles}\end{matrix} + \begin{matrix}\text{number of} \\ \text{noncolliding particles}\end{matrix}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Free paths depending on particle properties:

$$\lambda_{ave} = \frac{1}{\pi\,(r_1 + r_2)^2\,N} \cdot \frac{v_{ave}}{\sqrt{{v_{1ave}}^2 + {v_{2ave}}^2}}$$

$r1$: radius ofthe 1st particle

$r2$: radius ofthe 2nd particle

v_{1ave}: average speed of the 1st particle

v_{2ave}: average speed of the 2nd particle

N: number of particles in unit volume (1 cm^3)

1 DISCHARGE THEOREMS

1. Townsend Theorem

Free paths depending on particle properties:
Molecule-molecule collision – mean free path of molecules:

$$r_1 = r_2 = r$$

$$v_{1ave} = v_{2ave} = v_{ave}$$

$$\lambda_{m,ave} = \frac{1}{\pi\,(2r)^2\,N} \cdot \frac{v_{ave}}{\sqrt{{v_{ave}}^2 + {va_{ve}}^2}}$$

$$= \frac{1}{4\sqrt{2}\pi r^2 . N}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Free paths depending on particle properties:
Molecule-ion collision – mean free path of ion:

$$r_{ion} \cong r_{molecule} \cong r$$

$$v_{ion} > v_{molecule} \rightarrow ignore\ v_{molecule}$$

$$\lambda_{i,ave} = \frac{1}{\pi\,(2r)^2\,N} \cdot \frac{v_{ion}}{\sqrt{{v_{ion}}^2}} = \frac{1}{4\pi r^2 . N}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Free paths depending on particle properties:
Molecule-electron collision – mean free path of electrons:

$$r_{electron} \ll r_{molecule} = r$$

$$v_{electron} \gg v_{molecule} \rightarrow ignore\ v_{molecule}$$

$$\lambda_{e,ave} = \frac{1}{\pi\, r^2\, N} \cdot \frac{v_{electron}}{\sqrt{{v_{electron}}^2}} = \frac{1}{\pi r^2 . N}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Free paths depending on particle properties:
Comparison of mean free path of electron, ion and molecule:

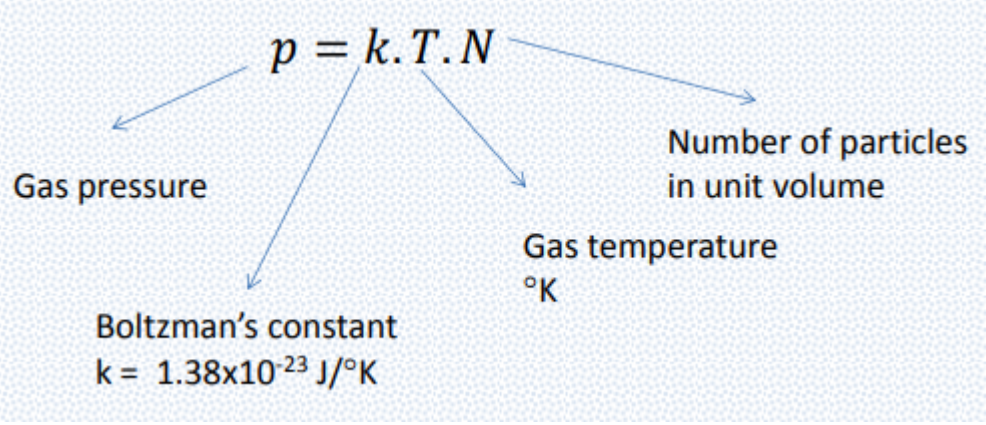
$$\lambda_{e,ave} = 4\lambda_{i,ave} = 4\sqrt{2}\lambda_{m,ave}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Determination of Townsend's coefficient:

Due to gas general law:



1 DISCHARGE THEOREMS

1. Townsend Theorem

Determination of Townsend's coefficient:

For an electron:

$$\lambda_{e,ave} = \frac{1}{\pi r^2 . N}$$

Substitute this $N = \frac{p}{k.T}$

$$\lambda_{e,ave} = \frac{1}{\pi r^2 . \frac{p}{k.T}} = \frac{k.T}{\pi r^2 . p} = f(T, p)$$

$\frac{k.T}{\pi r^2} = \frac{1}{A}$ Constant!!! (if T is constant)

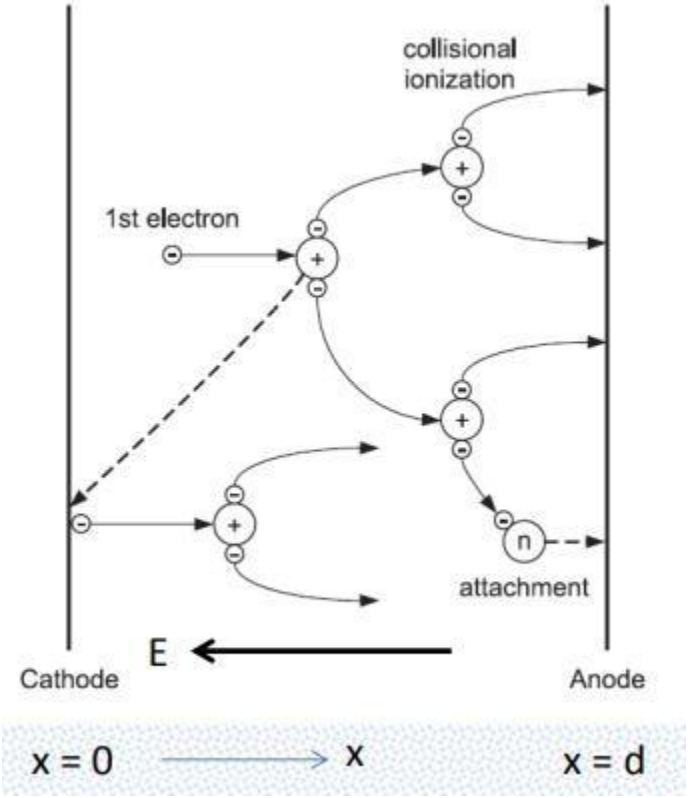
$$\lambda_{e,ave} = \frac{1}{A.p}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Townsend's first ionization coefficient, α:

It is the number of ionizing collisions, on average, made by one electron per unit drift (1 cm) in the direction of the field.



1 DISCHARGE THEOREMS

1. Townsend Theorem

Townsend's first ionization coefficient, α:

$$\alpha = \underbrace{\frac{1}{\lambda_{e,ave}}}_{\text{Number of collisions}} \underbrace{e^{-\frac{\lambda_{e,i}}{\lambda_{e,ave}}}}_{\text{Probability of ionization in one collisison}}$$

$$\lambda_{e,ave} = \frac{1}{A.p}$$

$$\lambda_{e,i} = \frac{V_i}{E}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Townsend's second ionization coefficient, γ

The first coefficient, α

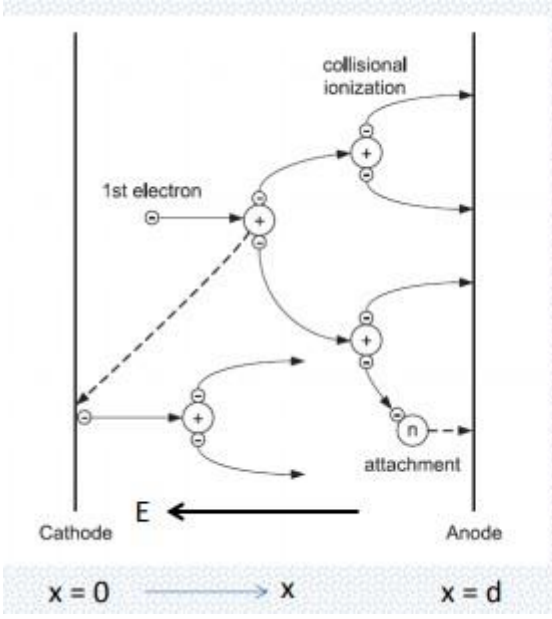
is the number of ion pairs made by one positive ion per unit drift (1 cm) in the direction of the field.

Positive ions removes electron from the cathode when they strike to cathode.

Townsend's ionization coefficient is about that phenomena, not about ionization by collision.

1 DISCHARGE THEOREMS

1. Townsend Theorem



γ: is the number of electrons removed from the cathode by one positive ion striking to cathode.

$$\gamma: \frac{1}{50}, \ldots, \frac{1}{5000}$$

If γ =1/50, one electron can be removed from the cathode when 50 positive ions strike to it.

1 DISCHARGE THEOREMS

1. Townsend Theorem

$\gamma = 0.02 \sim 0.00002$

γ depends on the material of the cathode and the type os gaseous insulator.

Material of cathode	Gas	
	Air	Neon
Al	0.035	0.1
Cu	0.025	0.065
Fe	0.02	0.06

1 DISCHARGE THEOREMS

1. Townsend Theorem

At a distance x = d,

$(e^{\alpha d} - 1)$ positive ions are produced by collison

$\gamma(e^{\alpha d} - 1)$ electrons are produced

Number of electrons removed from the cathode by one positive ion

1 DISCHARGE THEOREMS

1. Townsend Theorem

$$\gamma\left(e^{\alpha d}-1\right)=1$$

If this condition is fulfilled, at least one electron is removed from the cathode and discharge can continue.

This is also called as; **condition of self-sustainability of discharge.**

1 DISCHARGE THEOREMS

1. Townsend Theorem

Equation for breakdown voltage in uniform fields by Townsend's theory (Paschen Equation)

$$U_b = \frac{B.p.d}{\ln\left[\frac{A.p.d}{\ln(1+\frac{1}{\gamma})}\right]}$$

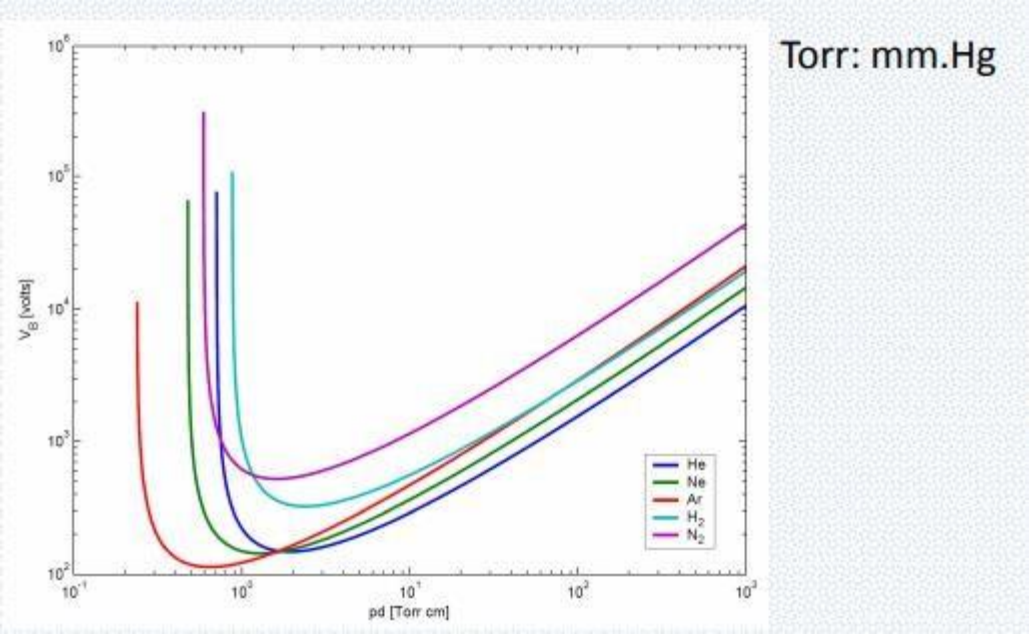
$$U_b = \mathrm{f(p.d)}$$

Breakdown voltage depends on multiplication of pressure and electrode gap

1 DISCHARGE THEOREMS

1. Townsend Theorem

Breakdown voltage curve (Paschen curve)



1 DISCHARGE THEOREMS

1. Townsend Theorem

The critical (p.d) value that makes the breakdown voltage minimum

$$(p.d)_{cr} = \frac{e.\ln(1+\frac{1}{\gamma})}{A}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

The critical (p.d) value that makes the breakdown voltage minimum

$$U_b[(p.d)_{cr}] = U_{b,min} = \frac{B.(p.d)_{cr}}{\ln\left[\frac{A.(p.d)_{cr}}{\ln(1+\frac{1}{\gamma})}\right]}$$

$$U_{b,min} = \frac{B.\frac{e.\ln(1+\frac{1}{\gamma})}{A}}{\ln\left[\frac{A.\frac{e.\ln(1+\frac{1}{\gamma})}{A}}{\ln(1+\frac{1}{\gamma})}\right]} = B.\underbrace{\frac{e.\ln(1+\frac{1}{\gamma})}{A}}_{(p.d)_{cr}}$$

$$U_{b,min} = B.(p.d)_{cr}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

The critical (p.d) value that makes the breakdown voltage minimum

$$E = {}^{U}/_{d}$$

$$E_b = {}^{U_b}/_{d}$$

$$E_b = \frac{B.p}{\ln\left[\frac{A.p.d}{\ln(1+\frac{1}{\gamma})}\right]}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

Generalized Pasched Equation

$A \rightarrow A.{}^{T}/_{T_0}$

$B \rightarrow B.{}^{T}/_{T_0}$

$$U_b = \frac{B.\frac{T}{T_0}.p.d}{\ln\left[\frac{A.\frac{T}{T_0}.p.d}{\ln(1+\frac{1}{\gamma})}\right]}$$

- $T_0 = 273 + 20°C = 293°K$
- $T = 273 + \theta°C$: gas temperature in $°K$
- p: gas pressure in mmHg
- A, B: gas constants
- γ: number of electrons removed from the cathode by positive ions

1 DISCHARGE THEOREMS

Streamer Theory

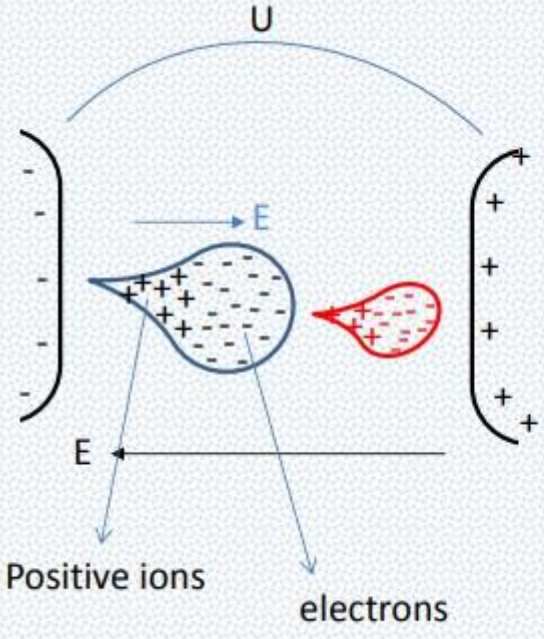
This theory explains electrical discharges in big gaps and high pressure.

p.d>500 mmHg.cm

The Streamer Theory predicts the development of a spark discharge directly from a single avalanche.

DISCHARGE THEOREMS

Streamer Theory



the electrons advance rapidly, the positive ions are left behind in a relatively slow-moving tail

Electric field between the avalanche and the anode is big

DISCHARGE THEOREMS

Streamer process

• The space charge produced in the avalanche causes sufficient distortion of the electric field that those free electrons move towards the avalanche head, and in so doing generate further avalanches in a process that rapidly becomes cumulative.
• As the electrons advance rapidly, the positive ions are left behind in a relatively slow-moving tail
• The field will be enhanced in front of the head. Just behind the head the field between the electrons and the positive ions is in the opposite direction to the applied field and hence the resultant field strength is less.
• Again between the tail and the cathode the field is enhanced.
• Due to the enhanced field between the head and the anode, the space charge increases, causing a further enhancement of the field around the anode.
• The process is very fast and the positive space charge extends to the cathode very rapidly resulting in the formation of a streamer.

DISCHARGE THEOREMS

Breakdown voltage by Streamer Theory

$$U_b = 5.27 \times 10^{-7} \frac{\alpha . d . e^{\alpha d}}{\sqrt{\frac{d}{p}}}$$

U_b: V

α: 1/cm

d: cm

p: mmHg

$\sqrt{\frac{d}{p}}$: $\sqrt{\text{cm}}$

d: electrode gap [cm] (electrode seperation)

p: gas pressure [mmHg]

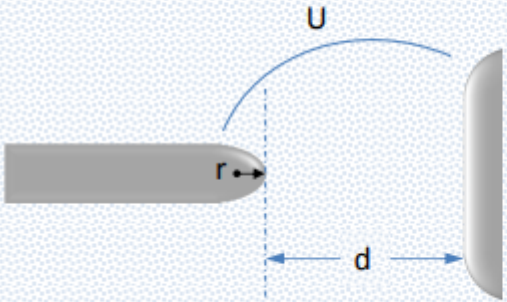
α: ionization coefficient of the eletron [1/cm]

U_b: breakdown voltage [V]

$$E_b = \frac{U_b}{d} = 5.27 \times 10^{-7} \frac{\alpha . e^{\alpha d}}{\sqrt{\frac{d}{p}}}$$

DISCHARGE THEOREMS

Streamer Theory for non-uniform electric field



r: curvature radius

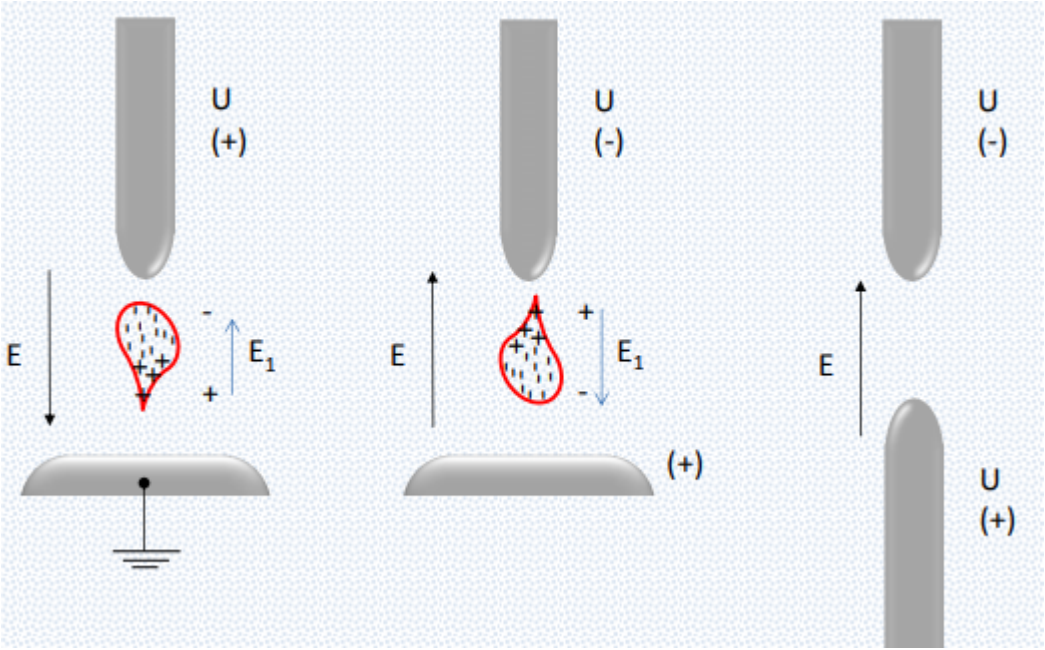
d: electrode gap

Non-uniform field:

$$\frac{d}{r} > 40 \rightarrow d > 40\,r$$

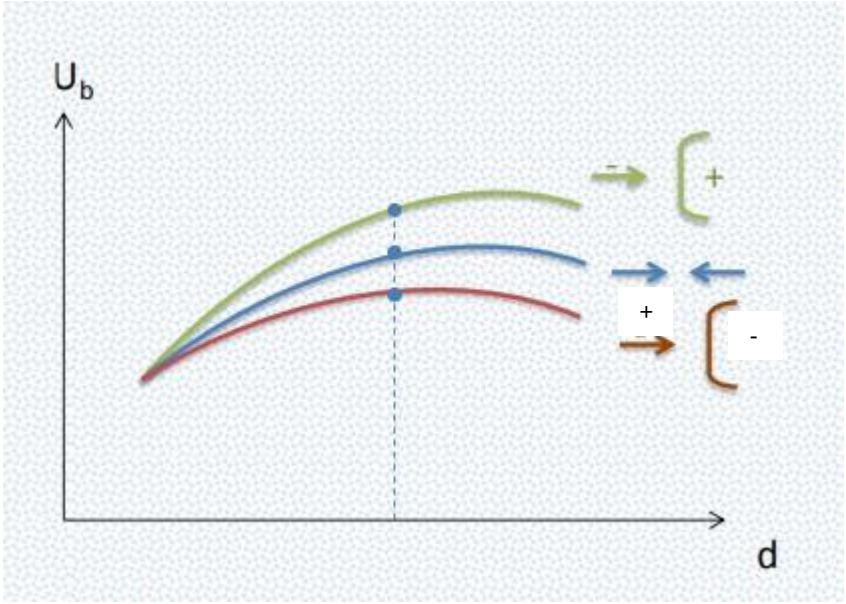
DISCHARGE THEOREMS

Different non-uniform electric field cases



DISCHARGE THEOREMS

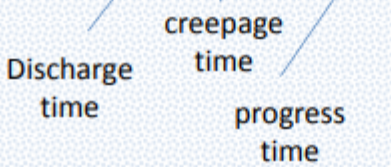
Electrode gap – breakdown voltage



1 DISCHARGE THEOREMS

Discharge time

$$t_b = t_\sigma + t_p$$



Creepage time: time required for an electron to appear after the moment which voltage applied

Progress time: time required for such an electron appeared in the gap to lead a breakdown after the moment it appeared